<SAR ADC>

Team members:

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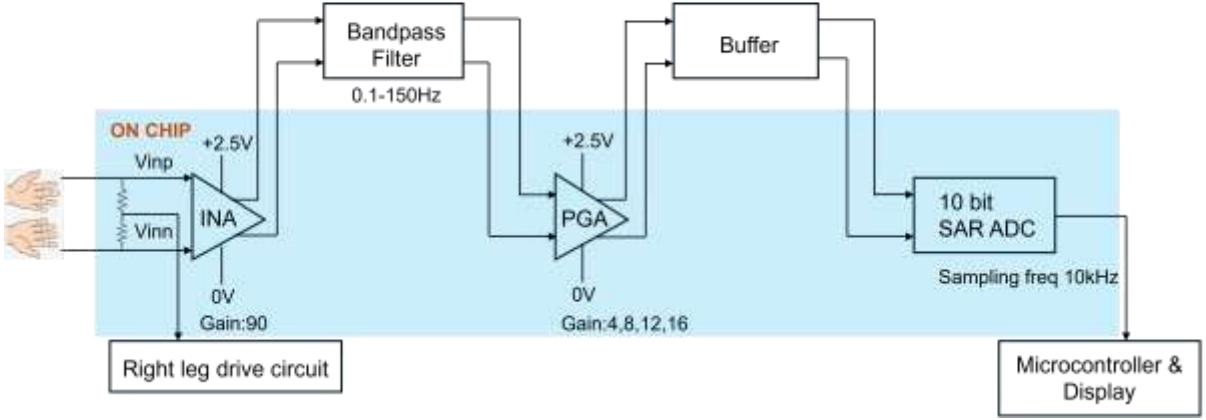
Project information

- Goal

  The goal of this project is to develop a low-power integrated ECG acquisition system that accurately amplifies, conditions, and digitizes cardiac signals using a fully differential SAR ADC.

- Design - High Level Proposal
  - ECG AFE (INA + BPF + PGA + Buffer) → Fully Differential
  - 10-bit SAR ADC (Bottom-Plate Sampling + Synchronous SAR Logic)



- Estimated Core Area
  - ECG Analog Front-End: ~0.3–0.6 mm²
  - SAR ADC and Digital Logic: ~0.2–0.4 mm²
  - Total Core Area: ~0.5–1.0 mm²
- Block Size Estimate
  - Proposed Core Size: 1 mm × 1 mm
  - Recommended Full-Chip Size with Pads: 2 mm × 2 mm

Project information

- Application

  Wearable and portable ECG monitoring systems for continuous cardiac signal acquisition and digitization.

- References

[1] B. Razavi, Design of Analog CMOS Integrated Circuits.
[2] R. Jacob Baker, CMOS Circuit Design, Layout, and Simulation.
[3] Tony Chan Carusone, Analog Integrated Circuit Design.

Team background

- Academic Experience

  - Relevant courses: Analog Electronic Circuits, Digital VLSI Circuits, Power Management Integrated Circuits, Advanced Analog Integrated Circuits, Analog-Digital Interfaces in VLSI, Microwave Circuit Design
  - Project experience: Transistor-level circuit design of DAC, LDO, OTA, LNA, buck converter, and fully custom microcontroller core

Questions, Suggestions, Doubts?

- Is the proposed project scope appropriate for the project timeline and expectations?
- Would you recommend implementing an 8-bit or a 10-bit SAR ADC considering the trade-off between complexity and performance?
- We are currently planning to implement a synchronous SAR ADC. However, if time permits, we may explore an asynchronous SAR architecture.